Student Performance & Placement Prediction

This project analyzes student academic behavior and develops machine learning models to predict exam performance and placement outcomes.

The project aims to:

• Understand factors influencing student success

• Predict final exam scores

• Predict placement outcomes

• Generate actionable educational recommendations

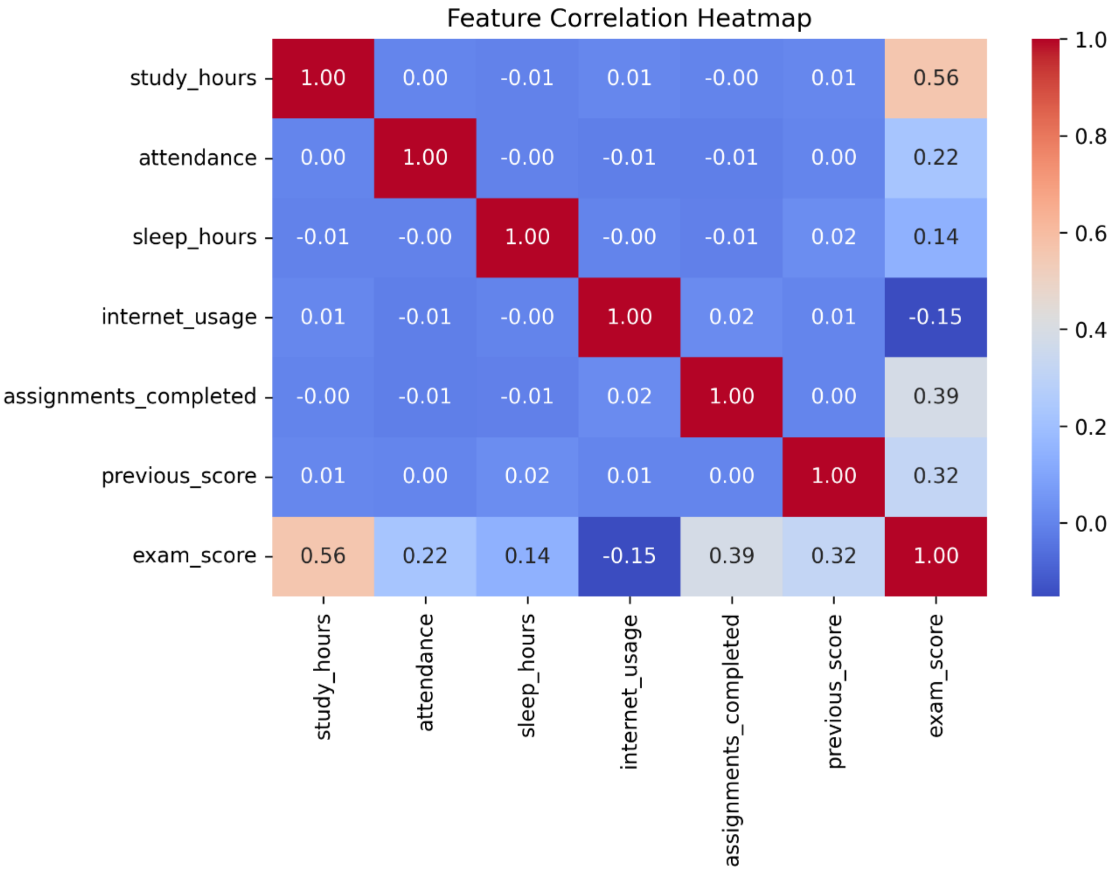
Source: Student Performance Prediction Dataset

Records: 10,000

Features:

- Study Hours
- Attendance
- Sleep Hours
- Internet Usage
- Assignments Completed
- Previous Score
- Exam Score
- Placement Status

Exploratory Data Analysis



Study Hours showed the strongest relationship with exam performance (0.56 correlation).

Assignments Completed and Previous Score also demonstrated meaningful relationships.

Internet Usage showed a weak negative relationship with exam performance.

Placement Distribution

Placed: 83.6%

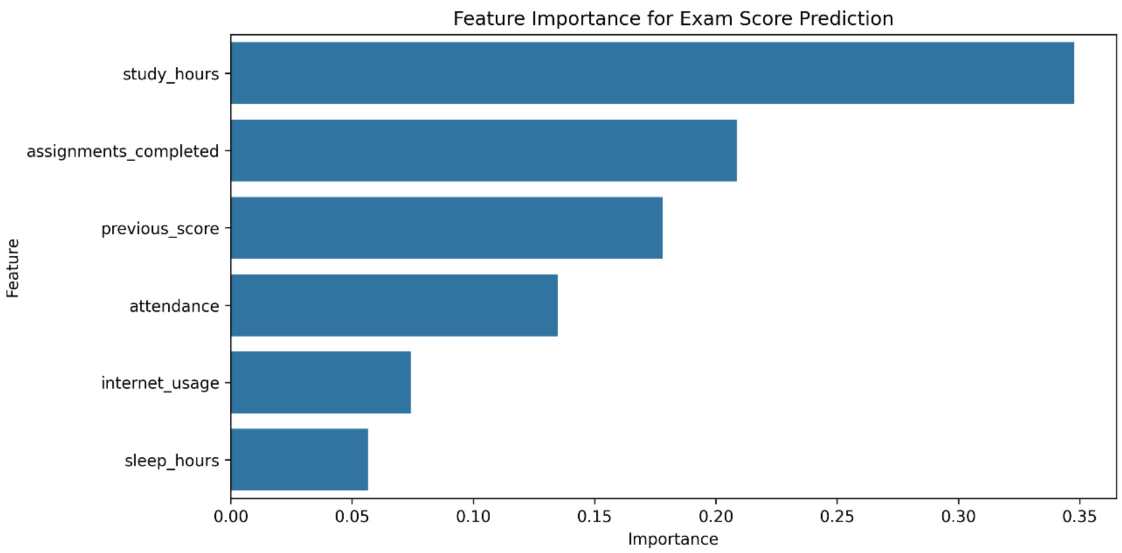
Not Placed: 16.4%

The dataset is imbalanced, requiring evaluation metrics beyond simple accuracy.

Machine Learning Results

Model	**R^2**	**MAE**
Linear Regression	0.663	6.92
Random Forest	0.713	5.88

Random Forest achieved the strongest performance, explaining approximately 71% of exam score variance.



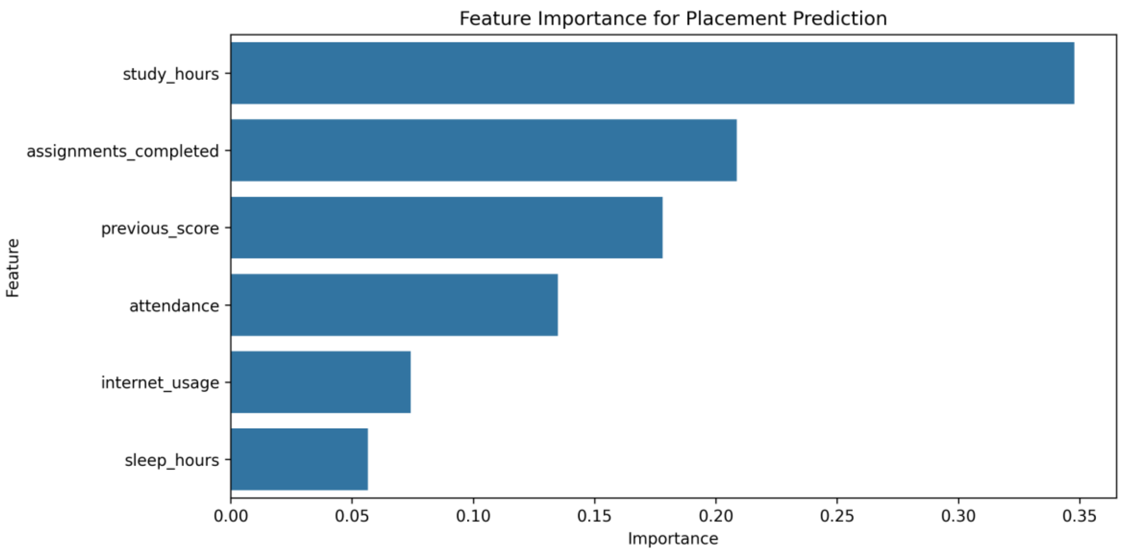
Key Predictors

1. Study Hours
2. Assignments Completed
3. Previous Score
4. Attendance

Placement Prediction

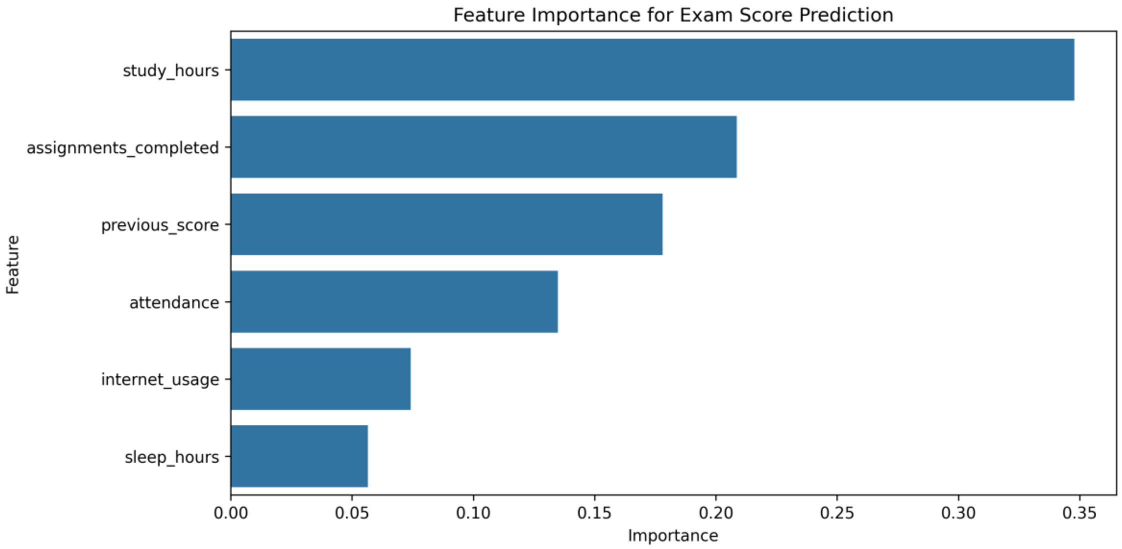
Model	**Accuracy**	**F1**
Logistic Regression	90.7%	94.5%
Random Forest	89.2%	93.7%

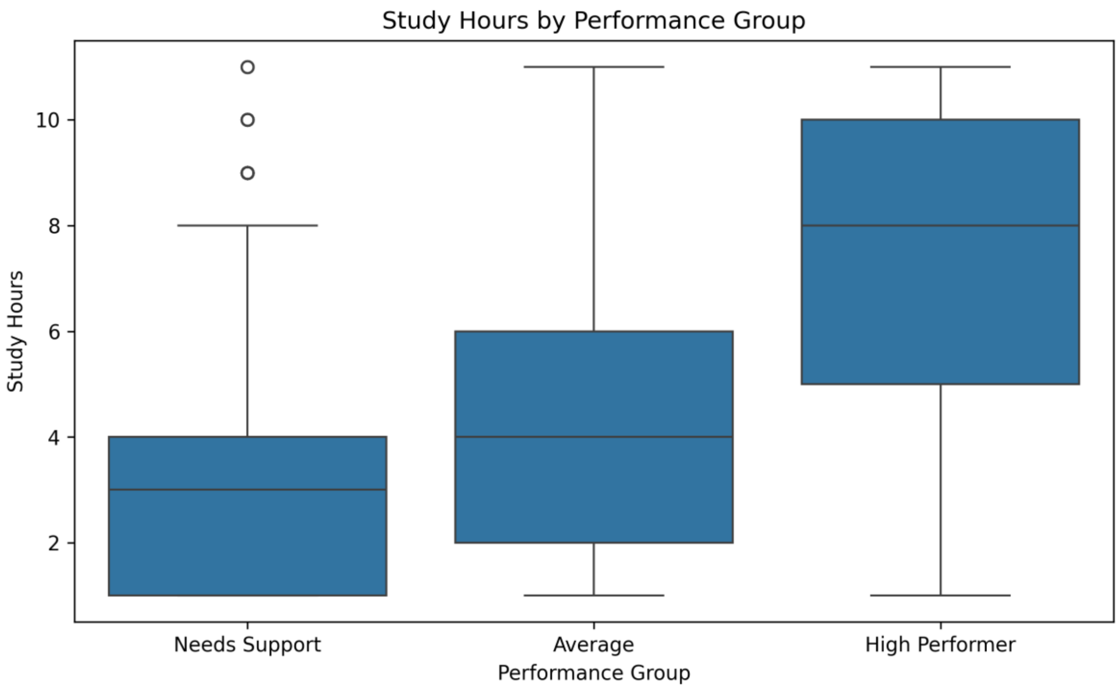
Logistic Regression slightly outperformed Random Forest, suggesting a relatively linear relationship between academic factors and placement outcomes.



Placement outcomes were strongly associated with study habits, assignment completion, previous academic performance, and attendance.

Student Success Insights & Recommendations





Student Success Characteristics

High-performing students consistently:

• Studied more hours

• Completed more assignments

• Maintained higher attendance

• Achieved stronger previous academic results

Educational institutions should:

1. Encourage consistent study habits.
2. Monitor assignment completion rates.
3. Identify students with declining attendance.
4. Provide early intervention for students with weak historical performance.

Conclusion

Academic engagement variables were consistently identified as the strongest predictors of student success.

Study hours, assignment completion, attendance, and previous academic achievement emerged as the primary drivers of both exam performance and placement outcomes.